Karnataka Residential Educational Institutions Society

ICT Level 3

Teaching Guide

Level: Level 3

Class: Classes 8, 9, 10

Duration: June - January (8 months)

Period: 40 minutes

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Curriculum Overview

ICT Level 3 covers advanced topics for Classes 8 through 10. The same textbook is used across all three classes, with differentiated instruction tailored to the developmental stage and academic expectations of each class.

Themes Covered

Chapter	Periods	Key Topics
Ch 1: Advanced Programming Concepts	12	Functions, Arrays, String operations, Nested loops
Ch 2: Data Structures	12	Lists, Dictionaries, Tuples
Ch 3: Database Fundamentals	12	Database concepts, SQL basics
Ch 4: Web Technologies	12	HTML basics, CSS introduction
Ch 5: Algorithms and Computational Thinking	12	Algorithm design, Pseudocode
Ch 6: Emerging Technologies	12	AI basics, IoT, Cloud computing

Differentiated Pedagogy by Class

Class 8 (Foundation)

Class 8 students require heavy Concrete-Phase-Abstract (C-P-A) scaffolding and guided practice. Instruction should rely on visual aids, step-by-step demonstrations, and structured worksheets. Peer collaboration is encouraged through pair programming and group tasks. Concepts are introduced with real-world analogies before transitioning to abstract representations.

Class 9 (Application)

Class 9 students benefit from a balanced approach between guided and independent learning. They are expected to apply previously learned concepts to new contexts with decreasing levels of teacher scaffolding. Emphasis is placed on debugging skills, logical reasoning, and collaborative problem-solving. Students begin to work on mini-projects with defined parameters.

Class 10 (Mastery)

Class 10 students focus on independent problem-solving and board exam preparation. Instruction emphasizes revision, application of concepts under timed conditions, and analysis of past examination papers. Students are expected to work with minimal scaffolding and demonstrate mastery through capstone projects and self-directed inquiry.

5E Model Implementation

Phase	Implementation Strategy
Engage	Activate prior knowledge through demonstrations, thought-provoking questions, or short videos related to the topic. Present a real-world problem that motivates the lesson.
Explore	Students engage in hands-on activities such as coding exercises, database queries, or web page construction. Teacher facilitates but does not direct.
Explain	Teacher-led discussion clarifies key concepts, introduces terminology, and formalises rules observed during exploration. Students articulate their understanding.
Elaborate	Students apply learning to new scenarios, extend existing programs, or tackle extension tasks. Emphasis on transfer of knowledge.
Evaluate	Formative assessment through observation, questioning, peer review, and exit tickets. Summative assessment through chapter-end exercises and projects.

Lesson Plan Structure (40 Minutes)

Time	Phase	Activity
0-5 min	Engage	Hook activity: demonstration, question, or quick recall quiz
5-15 min	Explore	Hands-on coding or activity worksheet; students work individually or in pairs
15-25 min	Explain	Teacher clarifies concepts, introduces terminology, addresses misconceptions
25-35 min	Elaborate	Extension task or application problem; independent or group work
35-40 min	Evaluate	Exit ticket, quick quiz, or self-reflection prompt

Assessment Framework

Assessment Type	Weightage	Components
Formative	40 marks	Class participation, Worksheets, Homework, Quizzes, Projects
Summative	60 marks	Mid-term exam, End-of-term exam, Practical assessment
Total	280 marks	Continuous assessment across three terms; final grade derived from cumulative scores

Resources Required

- Computers with Python installed (minimum one per pair of students)
- Textbook: ICT Level 3 (as prescribed by the state board)
- Printed worksheets and activity sheets for each chapter
- Projector and screen for demonstrations
- Access to online coding environments as backup (e.g., Replit, Trinket)
- Chart paper, markers, and sticky notes for C-P-A activities
- Database software (SQLite or equivalent for Chapter 3)
- Web browsers with developer tools enabled (for Chapter 4)

Dormitory Tasks

Dormitory tasks are assigned for evening self-study in the boarding facility. These tasks are designed to reinforce the day's learning and promote independent practice.

- **Code tracing exercises**: Students trace through given code snippets and predict output without running the program.
- **Debugging worksheets**: Intentionally flawed programs are provided for students to identify and correct errors.
- **Concept mapping**: Students create visual maps connecting related concepts from the chapter.
- **Reading assignments**: Selected sections of the textbook are assigned for pre-reading before the next lesson.
- **Mini-projects**: Short guided projects (e.g., building a calculator, creating a simple web page) that consolidate multiple concepts.

Capstone Projects

Capstone projects are assigned at the end of each term. They require students to integrate knowledge from multiple chapters and demonstrate cumulative understanding.

Project Guidelines

- Projects must address a real-world problem or scenario relevant to the student's context.
- Students may work individually or in groups of up to three.

- Each project requires a written component documenting the design process, algorithm, and reflections.
- Projects are presented to the class in the final week of the term.

Sample Project Topics

- **Library Management System**: A console-based program using functions, lists, and basic database operations to manage book records.
- **Personal Portfolio Website**: A multi-page website using HTML and CSS showcasing student work and interests.
- **Traffic Signal Simulator**: A Python program using loops and conditional logic to simulate traffic light sequences.
- **Data Analysis Report**: Students collect survey data, store it in a database, and generate a summary report using SQL queries.